

ALGORITHM – FINANCIAL DATA ANALYSIS USING PYTHON

Pandas Data Analysis Program

Algorithm

1. Start the program.
2. Import the Pandas library using the statement: `import pandas as pd`.
3. Read the `financial_data.csv` file using `pd.read_csv()` and store the data in the variable `data`.
4. Display the first six rows of the dataset using `data.head()`.
5. Display the structure and information of the dataset using `data.info()`, including the number of entries, columns, data types, and memory usage.
6. Generate and display summary statistics using `data.describe()`, including count, mean, standard deviation, minimum, quartiles, and maximum.
7. Import the Tkinter library and the `askopenfilename` function from `tkinter.filedialog` if a file-selection window is required.
8. Hide the main Tkinter window using `Tk().withdraw()`.
9. Display the results in the Python/VS Code terminal.
10. Stop the program.

Output

The program displays the first rows of the dataset, dataset information, and descriptive summary statistics such as count, mean, standard deviation, minimum, quartiles, and maximum.

Program Screenshot

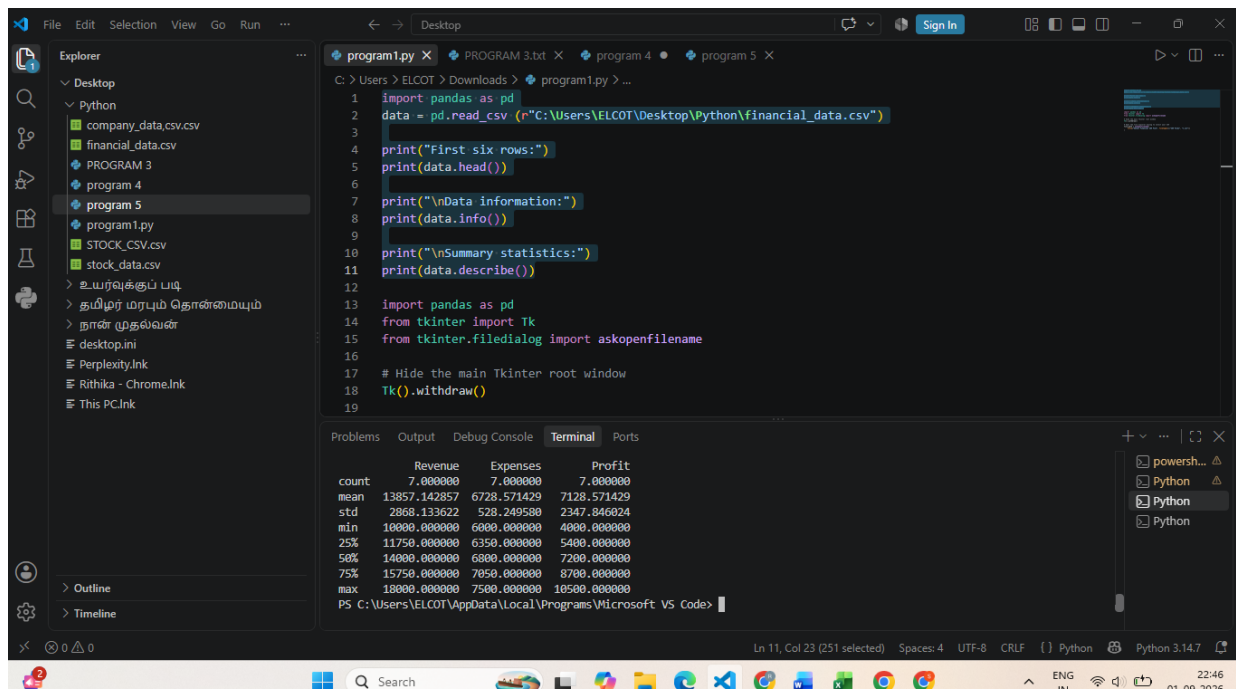


Figure 1: Python program and terminal output in Visual Studio Code